

SOC and Current-Sensing Graph Analysis

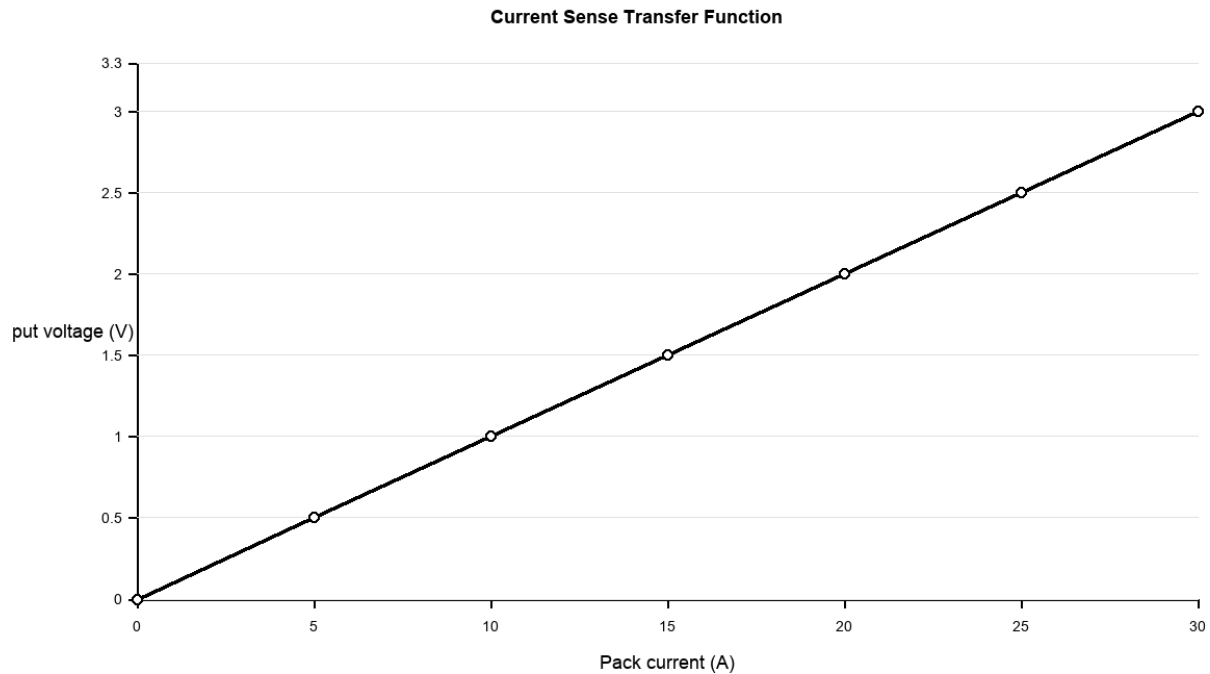
Battery Management System for Electric Scooter | ECE 445 Team 35 | Technical Design Note

Current-Sense Transfer Function

This design note graphs the current-sense path used by the master board. The pack current is converted to a shunt voltage by a 1 mOhm resistor and then amplified by the INA240 with gain of 100 V/V before being sampled by the STM32 ADC.

$$V_{\text{shunt}} = I_{\text{pack}} * R_{\text{shunt}}$$

$$V_{\text{ADC}} = G * V_{\text{shunt}} = 100 * I_{\text{pack}} * 0.001 \text{ Ohm}$$



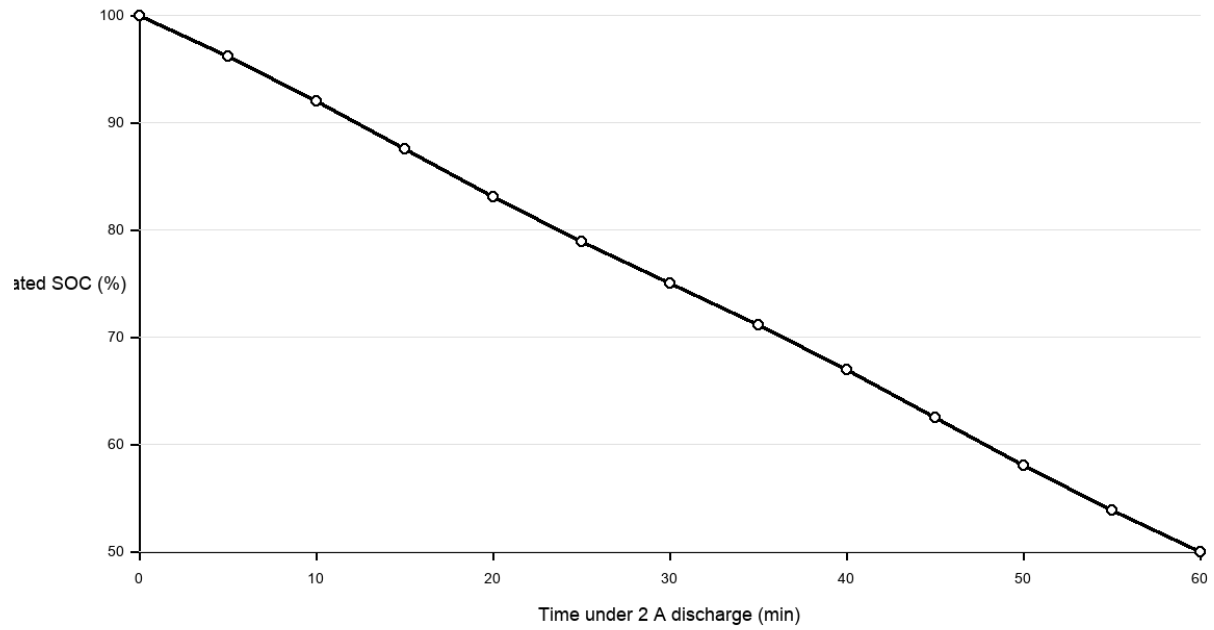
The plot shows that 30 A maps to approximately 3.0 V, which uses most of the 0 V to 3.3 V ADC range without saturating the input.

SOC Estimation Trend

SOC is estimated by integrating measured current over time. The graph below shows a representative 2 A constant-current discharge from a 4 Ah pack. Under ideal conditions, 2 A for 60 minutes removes 2 Ah, which is 50% of the nominal pack capacity.

$$\text{SOC}(t) = \text{SOC}(t_0) - (1 / Q_{\text{nom}}) * \int I(t) dt$$

Representative Coulomb-Counting SOC Trend



Quantity	Value	Interpretation
Q_nom	4 Ah	Nominal prototype pack capacity used for SOC scaling.
Discharge current	2 A	Representative load current for the plotted example.
Charge removed in 60 min	2 Ah	Equivalent to 50% of a 4 Ah pack.
Measured current check	2.01 A reference vs 1.97 A reported	Current error was below the +/-5% requirement.